YASHA SAXENA

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EDUCATION

Duke University, Durham, NC
PhD in Biomedical Engineering — Expected May 2027
Certificate in College Teaching
MS in Mechanical Engineering and Materials Science — Expected May 2027
Certificate in AI for Materials Science
MS in Biomedical Engineering — May 2016

Northwestern University, Evanston, IL
BA in Economics — June 2013
Minor in Mathematics, Business

ADDITIONAL CERTIFICATES

- **Professional Certificate in Digital Signal Processing**, UCSD Extension – 4.00/4.00 (March 2021)
- **Premedical Studies Post-Baccalaureate Program**, UC Berkeley Extension – 4.00/4.00 (July 2016)

TECHNICAL SKILLS

- *Computer Languages/Tools:* Python, R, MATLAB, C++, Git, PyTorch
- *Mechanical Design/Simulation:* OnShape, SolidWorks, Fusion360, Blender
- *Electronics Design/Testing:* KiCad, Altium Designer, Protoboarding, Oscilloscopes, Multimeters
- *Laboratory:* Mammalian Cell Culture, Hydrogel Fabrication, Polymer Synthesis, Microfluidics, Confocal Imaging, Immunofluorescence Staining

ACADEMIC EXPERIENCE

Duke University, Durham, NC — 08/2022 - Present

Graduate Research Assistant, Segura Lab (08/2022 - Present)

- Investigating the effects of material structure on cellular behavior in granular hydrogel wound healing materials
- Developing computational models for structure-function analysis in granular tissue engineering materials
- Mentoring undergraduates on independent study biomaterials research projects
- Developed a computational approach to characterize and identify 3D pore shapes in granular tissue engineering materials
- Developed an inverse design model to select granular hydrogel design parameters for specific drug release kinetics

Teaching Assistant, BME 254L MedTech Prototyping (08/2026 - Present)

- Assisting students with assignments and laboratory work related to CAD design, printed circuit board fabrication, and medical device prototyping, testing and assembly
- Grading and troubleshooting engineering assignments
- Providing technical support with medical device projects during class and laboratory sections

Teaching Assistant, BME 728-729S BME TA Seminar I & II (08/2024 – 12/2025)

- Co-led a professional-development seminar for first- and second-time Biomedical Engineering teaching assistants, guiding students through teaching responsibilities, requirements, and expectations
- Designed the course syllabus and organized weekly sessions, including guest lectures and structured, discussion-based seminars focused on challenges encountered during early teaching experiences
- Reviewed and responded to students' weekly reflections on their teaching experiences, providing individualized guidance and feedback

- Managed weekly meal planning and ordering within an allocated budget to support student engagement during end-of-day seminar meetings

Teaching Assistant, BME 590L Machine Learning in Pharmacology (08/2023 - Present)
- Supported students in processing pharmacological datasets and developing machine-learning algorithms to predict chemical structures for drug discovery
- Evaluated students' problem-set Jupyter notebooks, providing feedback on code functionality, analytical approach, and consistency between outputs and expected results
- Assessed final projects in which students proposed and implemented original machine-learning models to address pharmacological research questions

Teaching Assistant, BME 244L Quantitative Physiology with Biostatistical Applications (08/2022 - Present)
- Taught foundational statistical concepts during weekly discussion sections, preparing students, to analyze physiological data collected in laboratory exercises
- Supported laboratory activities involving the collection of physiological data from human participants and ex vivo specimens using data acquisition equipment
- Evaluated detailed laboratory reports using comprehensive rubrics developed in collaboration with the course instructor and teaching team
- Advised students during office hours on statistical concepts, problem sets, and data-analysis approaches

INDUSTRY EXPERIENCE

Kardigan Bio, *Translational Research Intern*, South San Francisco, CA 05/2026 – 08/2026
- Tested a drug for calcified aortic valve stenosis through a disease model system with heart valve cells seeded in a PEG, collagen hydrogel
- Designed and 3D printed a test fixture to be able to image 18 gels simultaneously in a micro-CT scanner reducing gel imaging time by 6 hours
- Quantified micron-scale calcification on the hydrogels through a microCT image analysis pipeline

CoVar, *Machine Learning Engineer Intern*, Durham, NC 05/2025 – 08/2025
- Developed computer vision and machine learning models for time-resolved 3D reconstruction and classification of aerial objects (4D aerial objects)
- Applied Hidden Markov Models (HMMs) to animated 3D CAD data to classify deformable aerial objects across varying geometric and environmental conditions
- Created 3D animations in Blender to generate synthetic datasets for model training and validation
- Applied normalization and camera transformations to reconstruct geometrically accurate 4D point clouds from RGB-D video

Asensus Surgical, *R&D Product Development Engineer*, Durham, NC 01/2022 – 08/2022
- Prototyped early-stage surgeon console designs for a robotic laparoscopic surgical system
- Designed and implemented a hand-tracking fixture integrating Valve Index handheld controllers with ROS to capture hand pose during controller use
- Conducted an optical tracking study using OptiTrack cameras and markers to establish ground-truth measurements; developed a signal processing pipeline to evaluate latency and accuracy across multiple motion-tracking systems

Apple Inc, *Algorithm Quality Engineer*, Durham, NC 07/2019 – 12/2021
- Authored test plans and defined quantitative performance specifications for iPhone touch panel firmware algorithms
- Developed 50+ Python unit tests to evaluate touch panel image quality and algorithm performance
- Designed and executed an urgent user study within 5 days to validate a critical post-release firmware update supporting delivery before Thanksgiving

- Led an investigation into software-based mitigation of a hardware failure affecting ungrounded devices
- Created and maintained a weekly Tableau dashboard to visualize unit test results; served as the Human Interactiv Device Quality Engineering team's Tableau lead and won the Tableau@Apple data visualization competition

University of California San Francisco, *Medical Device Engineer*, San Francisco, CA 08/2018 – 06/2019
- Developed electronics for a force-sensitive resistor (FSR) pressure sensing mat to assess pressure ulcer risk in bedridden patients; reduced signal hysteresis error by 16% by switching from resistive to capacitive measurement
- Miniaturized a fluidic pressure monitor onto a 1-inch-square printed circuit board for blood pressure monitoring in a porcine artificial pancreas study
- Designed an IRB-approved clinical study with San Francisco General Hospital and UC Berkeley engineering students to collect emergency department patient data using an acoustic pneumonia detection prototype; authored a device troubleshooting manual for study moderators
- Developed a MySQL database hosted on Amazon Web Services (AWS) to organize clinical data for analysis

Baebies, *Systems Engineering Intern*, Durham, NC 01/2018 – 05/2018
- Performed verification and validation testing of optical sensors in a digital microfluidic platform for neonatal blood-based assays
- Developed Python tests to evaluate FINDER device hardware functionality, identifying a recurring DC shift in an Ocean Optics spectrometer after every 1000 reads
- Conducted statistical analyses of test results to support regulatory documentation

Genentech, *Pharma Technical Development Intern*, South San Francisco, CA 05/2017 – 08/2017
- Developed a QR code scanning system for laboratory reagents, reducing data entry time by 84%
- Designed a magnetic extraction tool to remove metal contaminants from assay beads during a heavy metal assay
- Performed colorimetric assays to detect endotoxins in drugs and devices

Workday, *UX Product Management Intern*, Pleasonton, CA 05/2016 – 08/2016
- Coordinated workflows between UX design and development teams across multiple software projects
- Created an internal documentation site explaining a backend infrastructure enhancement
- Conducted user experience research and competitive analysis to inform development of a new Workday product

Gensler, *Business Information Analyst,* Chicago, IL 01/2015 – 07/2015
- Designed and developed Tableau dashboards to visualize firm financial and productivity data; cleaned. And transformed underlying datasets using SQL
- Troubleshot Tableau technical issues and provided support to end users

Foote, Cone, & Belding, *Strategic Analytics Analyst*, Chicago, IL 11/2013 – 12/2014
- Analyzed consumer data to drive effective marketing and product development strategies for Fortune 500 companies
- Used Tableau and SQL to provide clients with interactive market research visualizations

PUBLICATIONS

- **Saxena, Y.***, Varberg, N., & Morton, K. (2026, June). Zero-shot classification of articulating 4D aerial objects through a Markovian synthetic template-matching approach. In *Synthetic Data for Artificial Intelligence and Machine Learning: Tools, Techniques, and Applications IV* (Vol. 14029, p. 82). SPIE.
- **Saxena, Y.***, Lin, P.-A., Shah, J., Asamoah, T., Jayaraman, A., Arya, G., & Segura, T. (2025). Factorial data-driven inverse design of granular hydrogels for targeted therapeutic release. *AI for Accelerated Materials Design Workshop, NeurIPS 2025*.

- **Saxena, Y**.*, Riley, L., Wu, R., Kabir, M. S., Randles, A., & Segura, T. (2025). 3D pore shape is predictable in randomly packed particle systems. *Matter*, *8*(12).